

Is autoimmunity the Achilles' heel of cancer immunotherapy?

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The emergence of immuno-oncology as the first broadly successful strategy for metastatic cancer will require clinicians to integrate this new pillar of medicine with chemotherapy, radiation, and targeted small-molecule compounds. Of equal importance is gaining an understanding of the limitations and toxicities of immunotherapy. Immunotherapy was initially perceived to be a relatively less toxic approach to cancer treatment than other available therapies-and surely it is, when compared to those. However, as the use of immunotherapy becomes more common, especially as first- and second-line treatments, immunotoxicity and autoimmunity are emerging as the Achilles' heel of immunotherapy. In this Perspective, we discuss evidence that the occurrence of immunotoxicity bodes well for the patient, and describe mechanisms that might be related to the induction of autoimmunity. We then explore approaches to limit immunotoxicity, and discuss the future directions of research and reporting that are needed to diminish it.